

4. Businesses are also going to cloud services providers for their big data solution requirements. This requirement falls into three broad categories, which are termed as the 'A-B-C' of big data. 'Analytical' solutions and high 'Bandwidth' solutions (required in the case of high-performance computing) comprise the first two requirements. In high Bandwidth area, NetApp has joined hands with open source vendors like Luster to create distributed a file system, to be deployed in a high-performance computing environment. The 'C' in the 'ABC' stands for large 'content' repositories. Projects like the UID or NPR, that need to deploy huge solutions that manage the data of one billion people, including their finger prints and iris scans, have to be stored in a single location. This data will be utilised by banks, security agencies and other branches of the government. So, managing and securing that kind of a content repository, which will continue to grow over a period of time, is extremely important. That's where Hadoop-based solutions play a crucial role.

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### Open source technology has an important role to play

Since data is growing at an exponential rate, the next stage would be where customers require high bandwidth solutions, both from the compute side as well as from the storage perspective. Those requirements may be for a specific period of time. Awasthi says, "One has to see whether one wants to fulfill those requirements from within one's own data centre or look at cloud services providers. That's where we are trying to work and provide solutions with the help of open source technology."

### Hadoop will be the winner!

Hadoop has emerged as a really important component. As the size of the content repository increases, the

requirement would be to scale the data storage across multiple systems and create a single distributed file system across all those storage systems. That is where the beauty of the Hadoop solution comes in, which brings in the HDFS file system over and above the different storage systems that are deployed by the customer. Awasthi says, "We have an example of how a Hadoop—based solution has helped NetApp in customer services. NetApp offers continuous monitoring of NetApp storage systems deployed at the customer site using 'Auto support', where we take direct feeds from all the systems which are enabled with 'Auto support'. This is done to give pro-active support to customers by doing predictive analysis on near online data received from various customers' systems. That's a huge volume of data, because we have multiple systems deployed across the world and each one of them is generating alerts or the other at any given point in time. When that information comes to our 'Auto Support' system, it creates a huge repository. Earlier it used to take months to create a report from a system on the huge data repository. Recently, we replaced our storage solution with our own Hadoop-based E-series solution at the back-end. The same work that used to take months is now done in a couple of hours or days. That is the benefit of open source technology—something we see within our own internal IT system. I am sure it wouldn't be different in any other business. For example, if a retail chain wants information about customer behaviour from various branches spread across the country, it's a major task to co-relate the vast data that is received from different sources. If the co-relation has to be done in a specified time to make a quick business decision, Hadoop plays a major role."

There are no security related fears in adopting open source technology either. In fact, the cloud services providers are happily adopting open source technologies because they help cut costs. Awasthi says, "I have seen a large number of telecom service providers as well as cloud services providers that look at open source based stacks to deploy cloud solutions. There is an alliance in the industry called the 'OpenStack', which does joint work in creating cloud environments using open source technologies. So, within India too, we have seen cloud service providers deploying solutions based on the open stack. People are more receptive to these kinds of solutions, provided they add value to their end customers. They are ready to implement these solutions in their infrastructure. Because in the end, the customer looks at what is the service level agreement (SLA) that the cloud service provider is ready to sign. They are least bothered about what technology is being used. Cloud services providers are looking at the way open source is spreading across data centres, and they find this to be a financially beneficial solution that can be deployed, while giving the customers the desired SLAs." 